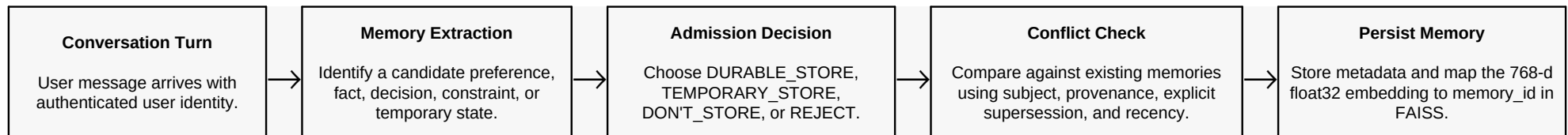


D4 - Conversational Memory System Data Flow

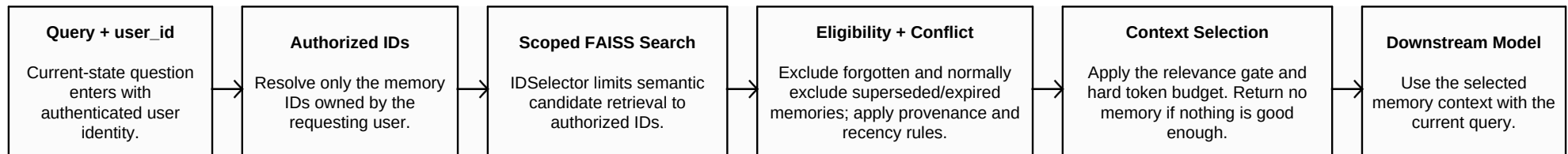
This document shows the data flows established during D4. It stays at the logical-design level and does not assume production infrastructure that has not been built.

1. Memory Creation / Update Flow



Important branch: REJECT and DON'T_STORE stop before durable memory. TEMPORARY_STORE stays scoped to its temporary use and should not silently become a durable preference.

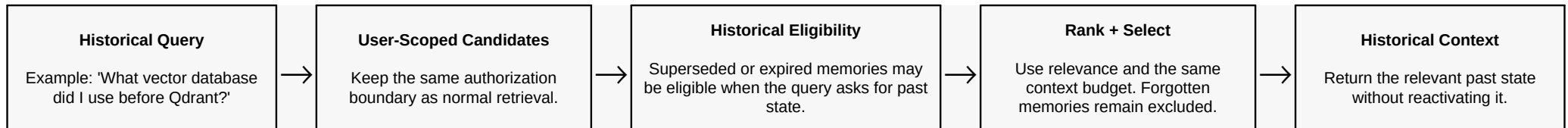
2. Current-State Retrieval Flow



Current-state rule: lifecycle status alone does not choose the answer. Authorization, query intent, relevance, provenance, explicit supersession, and recency all matter.

D4 - Historical, Lifecycle, and Forgetting Flows

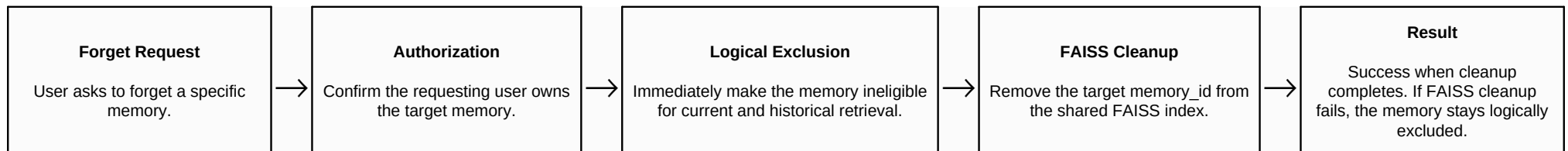
3. Historical Retrieval Flow



4. Lifecycle Update Flow

Trigger	State change	Result
Explicit replacement/correction	old active -> superseded; new memory -> active	Old state remains available for historical retrieval.
Explicit valid_until passes	active -> expired	Not treated as current by default.
Old value becomes current again	create a new active memory	Do not reactivate the old superseded record.

5. Forgetting Flow



Boundary: This flow reflects the D4 deletion design that was actually tested. Complete persistent crash recovery, database transactions, API transport, and production background workers are not assumed here.